SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

QUESTIONS 10 / 10 completed

Fib series-iterative

FIB SERIES-ITERATIVE & RECUR

Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

Sample Input:

PROGRAM EDITOR

C

Run Save

```c
#include<stdio.h>
#include<time.h>
int calls=0;
int fib(int n)
{
  calls++;
  if(n<=1)
    return n;
  return fib(n-1)+ fib(n-2);
}
int main()
{
  int n,i,a=0,b=1,c;
  clock_t start,end;
  scanf("%d",&n);
  printf("iterative: ");
  start=clock();
  for(i=0;i<n;i++)
  {
    if(i==0)
      printf("0 ");
```

OUTPUT

```
iterative: 0 1 1 2 3 5 8
recursive: 0 1 1 2 3 5 8
recurisive calls:59
iterative time 0.000 ms
recursive time:0.000 ms
```

INPUT

7

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Write a C program to generate Fibonacci series up to N terms using both iterative and recursive methods. The program should display both outputs and compare execution time in milliseconds. Also display total recursive calls made.

Sample Input:

Enter N: 7

PROGRAM EDITOR

C

Run Save

```
    else if(i==1)
      printf("1 ");
    else{
      c=a+b;
      printf("%d ",c);
      a=b;
      b=c;
    }
  }
  end=clock();
  printf("\n");
  printf("recursive: ");
  calls=0;
  start=clock();
  for(i=0;i<n;i++)
    printf("%d ",fib(i));
  end=clock();
  printf("\nrecurisive calls:%d",calls);
  printf("\niterative time %.3f ms",0.0);
  printf("\nrecursive time:%.3f ms",(double)(end-start)*1000/CLOCKS_PER_SEC);
  return 0;
```

OUTPUT

```
iterative: 0 1 1 2 3 5 8
recursive: 0 1 1 2 3 5 8
recurisive calls:59
iterative time 0.000 ms
recursive time:0.000 ms
```

INPUT

7